

A DIGITAL TWIN APPROACH TO ENERGY EFFICIENCY AND OCCUPANCY MONITORING

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Introduction

- Global warming and resource depletion demand urgent sustainable energy practices in buildings.
- Digital Twins enable real-time, adaptive energy solutions beyond static simulations.
- Home-Assistant is an open-source platform for building virtual replicas using IoT sensors.
- This work integrates DT with real-time data, ML occupancy detection, and personalized recommendations, validated in a lab.

Problem Statement

- Existing DTs are simulation-oriented and lack real-time implementation with live sensor feedback.
- Security and privacy are often overlooked, exposing sensitive building data to cyber threats.
- Occupancy detection relies on intrusive cameras or static datasets, not real-time environmental data.
- Need for a privacy-conscious, real-time DT unifying visualization, detection, and recommendations.

Objectives

- Build a real-time, data-driven Digital Twin on Home-Assistant that adapts to occupancy and environmental changes.
- Deploy on Raspberry Pi 4B+ for low-latency, privacy-preserving edge processing.
- Implement ML-based occupancy detection using non-intrusive ambient sensors (T, H, L, motion, sound, CO₂).
- Deliver personalized energy-saving recommendations and validate through real-world lab testing.

Literature Review

Author (Year)	Paper	Contribution	Limitation
Tan et al. [1]	<i>DT-driven indoor lighting via computer vision & dynamic BIM</i>	Fused computer vision and BIM for a visually-oriented DT lighting O&M platform with occupancy-based control	Relies on video-camera surveillance – intrusive, privacy concerns
Saad et al. [2]	<i>IoT-based DT for Energy Cyber-Physical Systems</i>	Modeled high/low-bandwidth DTs hosted on AWS, validated with real-time power/control measurements	Cloud-hosted, lacks local edge processing and occupancy detection
Li et al. [3]	<i>Data-driven hybrid Petri-net DT for manufacturing energy</i>	Real-time, data-driven energy behaviour model (DDHPN) for an energy-efficient manufacturing system	Domain-specific to manufacturing, not designed for residential use

Literature Review (cont.)

Author (Year)	Paper	Contribution	Limitation
Sardianos et al. [4]	<i>Room occupancy prediction from motion-sensor data</i>	Predicted occupancy using motion-sensor data with historical patterns to distinguish absence vs. inactivity	Relies solely on a single motion sensor – limited robustness
Sayed et al. [5]	<i>Edge-based real-time occupancy detection (non-intrusive)</i>	Deep-learning model on an edge device using ambient sensor data, 99.75% real-time accuracy	Evaluated only in a single university-office environment
Aliero et al. [6]	<i>Non-intrusive room occupancy prediction using ML</i>	Combined camera and environmental sensor data, large public dataset (40,000+ records), 99.70% accuracy	Still incorporates camera data, partially intrusive

Existing System

- Industrial DT platforms (MindSphere, Azure) target enterprise settings, not residential use.
- Camera-based occupancy is privacy-invasive, single PIR sensors are too simplistic.
- Some ML models use environmental data but depend on static, pre-existing datasets.
- No unified edge system combines real-time occupancy, sensing, and personalized recommendations.

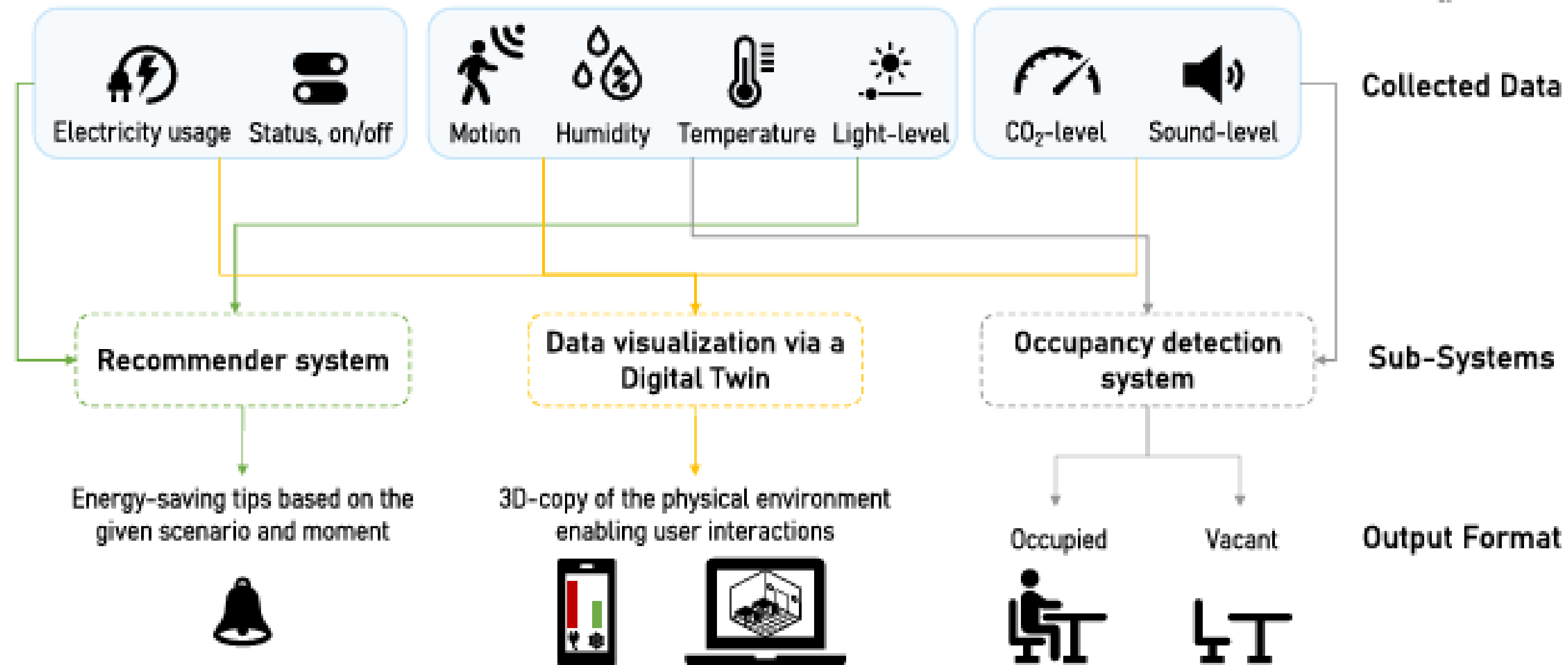
Limitations of Existing System

- DTs are used offline for forecasting, not real-time adaptive control with live data.
- Cloud dependency exposes sensitive data to cyber threats and privacy breaches.
- Camera-based detection is intrusive, ML models lack real-time, context-specific data.
- No integration between occupancy status, environmental conditions, and actionable energy tips.

Proposed Solution

- Build a lab-space DT on Home-Assistant with diverse IoT sensors and smart meters.
- Deploy on Raspberry Pi 4B+ for local, low-latency, privacy-preserving processing.
- ML occupancy detection using ambient data (T, H, L, motion, sound, CO₂).
- Deliver personalized energy tips via mobile and a 3D visual replica for immersive monitoring.

System Architecture



- Collected sensor data feeds three sub-systems: the Recommender system, DT Visualization, and Occupancy Detection.
- Each sub-system produces a distinct output format tailored to its purpose (tips, 3D view, occupied/vacant).

Algorithm 1: Implementation of Proposed Digital Twin Method

- Initialize DT: Set up Home-Assistant platform, configure sensors, and import environmental/occupancy datasets for real-time data collection.
- Data Collection & Preprocessing: Retrieve real-time sensor data (T, H, L, motion, sound, CO₂), clean, handle missing values, align timestamps, and update DT with current parameters.
- Occupancy Detection: Train ML model on historical labelled data, predict occupancy status (occupied/vacant) at each interval using latest readings and update DT accordingly.
- Recommender & Feedback: Generate energy-saving recommendations (turn off devices if vacant, open windows if external conditions favor natural cooling), send via mobile, collect user feedback, and refine model continuously.

Digital Twin Integration

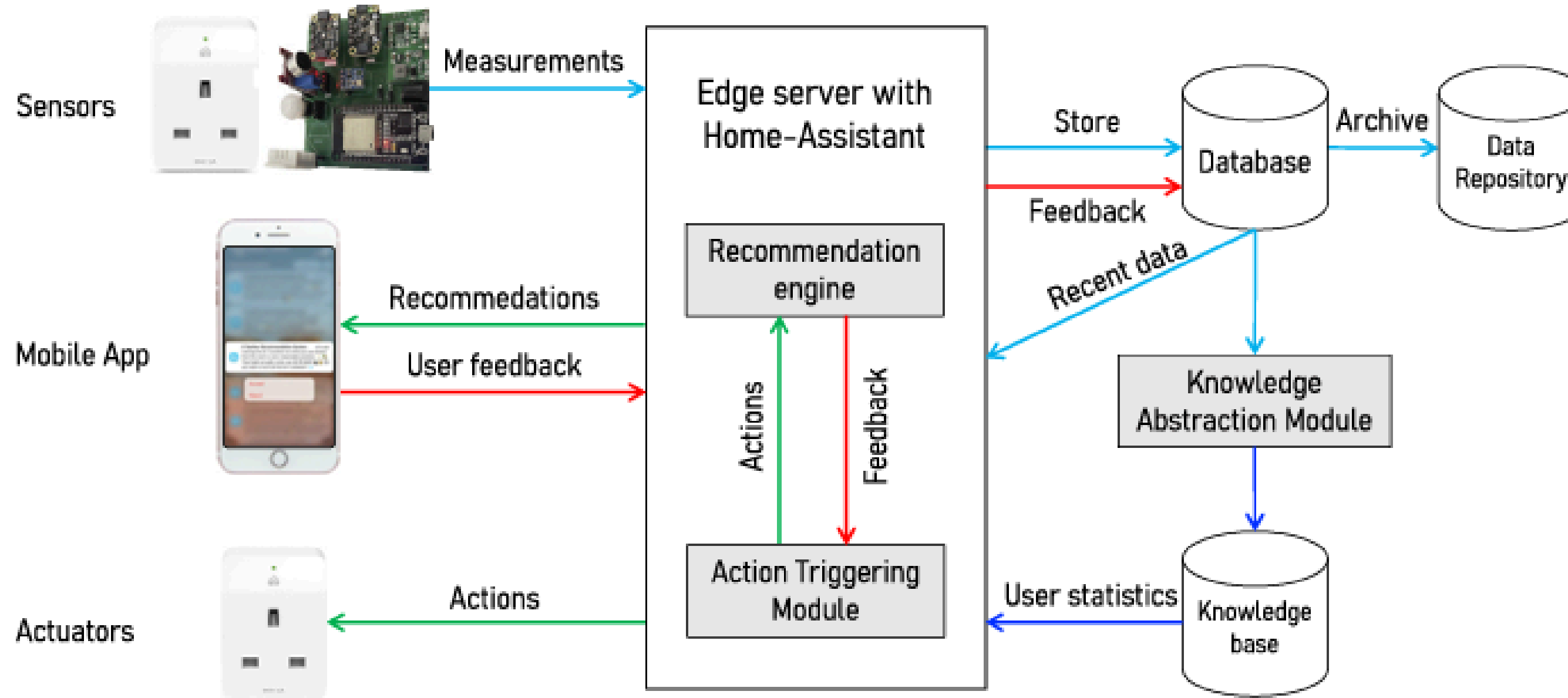
- Home-Assistant provides automation backbone with extensive add-ons and intuitive interface.
- Detailed 3D lab model built using Sweet Home 3D for interactive visualization.
- Home-Assistant runs on Raspberry Pi 4B+ as the edge device for local processing.
- Live sensor data syncs with the 3D model for dynamic, real-time visualization.



Occupancy Detection

- Six environmental features: temperature (T), humidity (H), luminosity (L), motion (M), sound (S), and CO₂ (C).
- Model learns $O = f(T, H, L, M, S, C)$ from historical labelled data (1=occupied, 0=vacant).
- Real-time prediction runs once per minute from latest sensor readings, updates 3D DT visual and triggers automations.
- Uses only real-time ambient data — no cameras or static datasets, distinguishing it from existing literature.

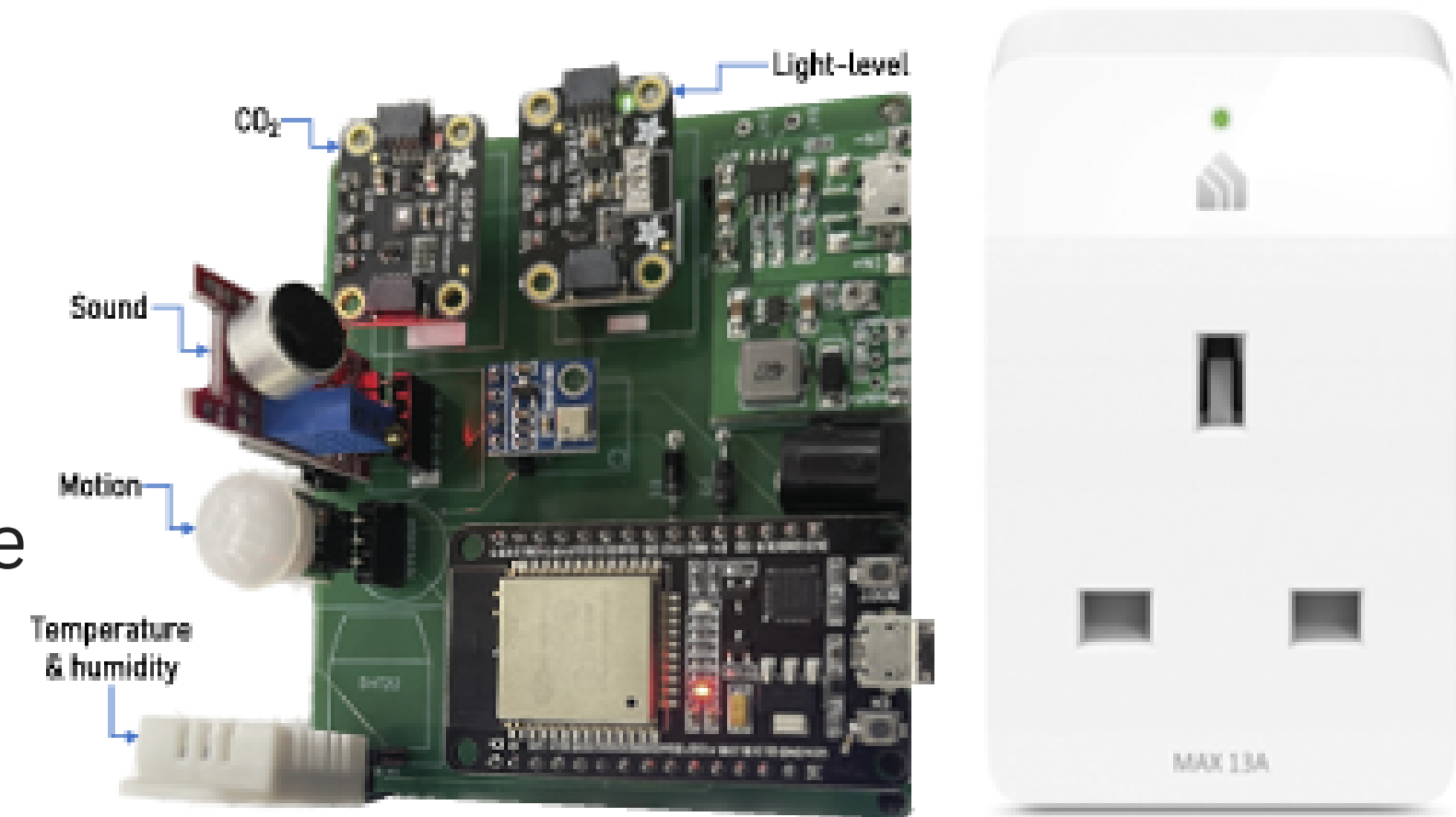
Recommender System



- The Action Triggering Module (ATM) receives real-time occupancy, appliance usage, and environmental data.
- Recommendations are delivered via mobile and Home-Assistant apps, and users can accept, reject, or disregard them.
- The system retrieves user preferences from a knowledge base and reviews recent sensor entries to select the best-suited message.
- Feedback is stored and used to refine future recommendations.

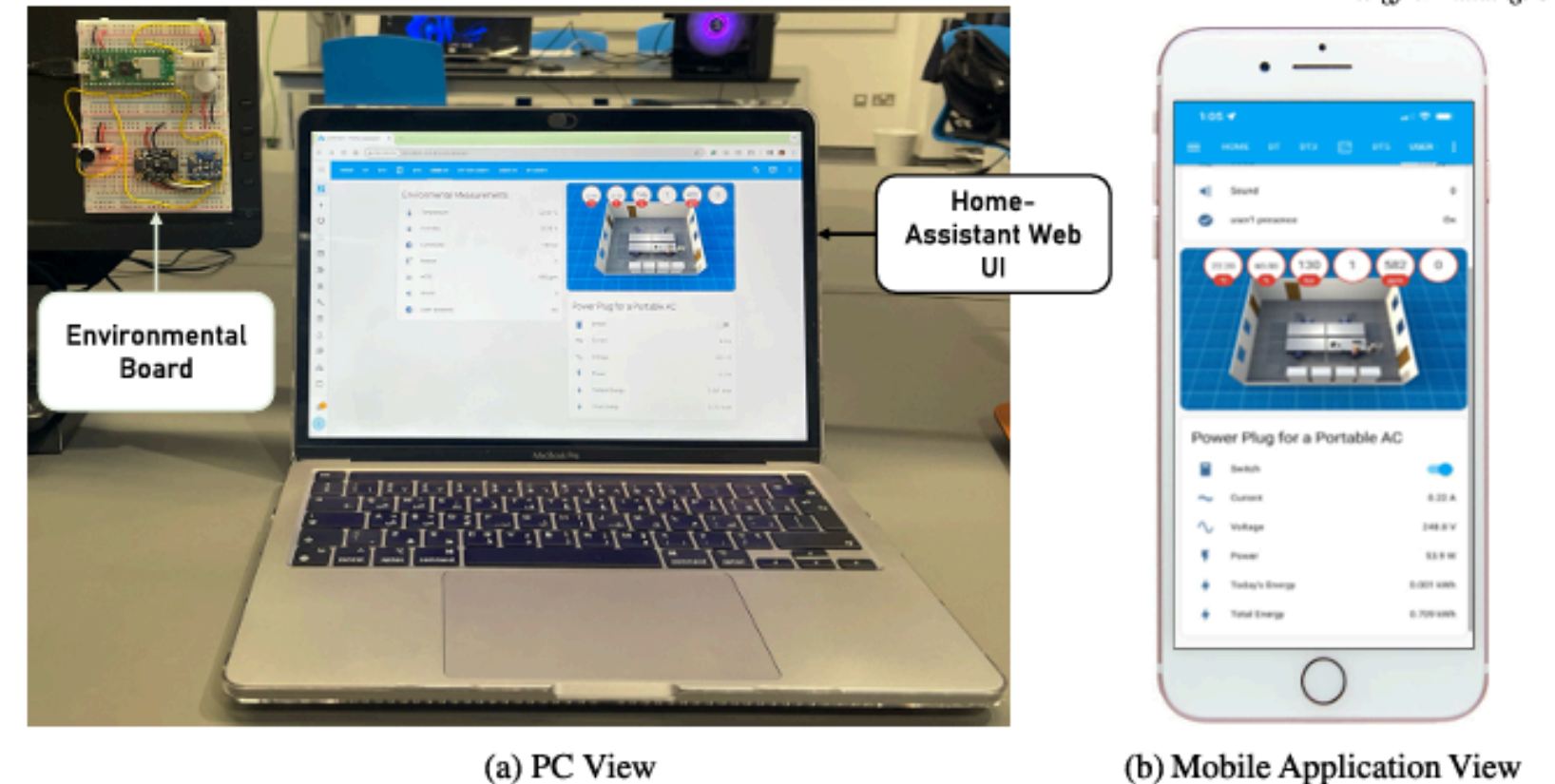
Sensor Data Retrieval

- Custom-designed environmental board collects temperature, humidity, luminosity, motion, sound, and CO₂ data.
- Off-the-shelf smart meter plugs measure appliance power consumption and enable remote on/off control.
- All data transmitted to Home-Assistant edge platform via lightweight MQTT publish-subscribe protocol.
- Fixed 10-second sampling interval ensures consistent, timely data collection and transmission.



Experimental Setup & Evaluation Metrics

- Controlled lab at ~22°C, all sensors calibrated, 7 days training + 3 days testing with real user, ground-truth occupancy manually logged.
- Classification metrics: Accuracy, Precision, Recall, and F1-score used to evaluate occupancy detection performance.
- Recommender performance measured via Acceptance Rate (% of recommendations users act upon).
- Energy savings calculated as: $\text{Reduction (\%)} = ((E_{\text{before}} - E_{\text{after}}) / E_{\text{before}}) \times 100$, comparing usage with vs. without recommendations.



Occupancy Detection Results



The Random Forest model closely tracks ground-truth occupancy across all three testing days (05–07 Feb 2024).

Occupancy Detection Results — Analysis

Occupancy state #

Un-occupied

0

748 52.9%	37 2.6%
32 2.3%	598 42.3%

Predicted class

0

1

Target class

95.12%

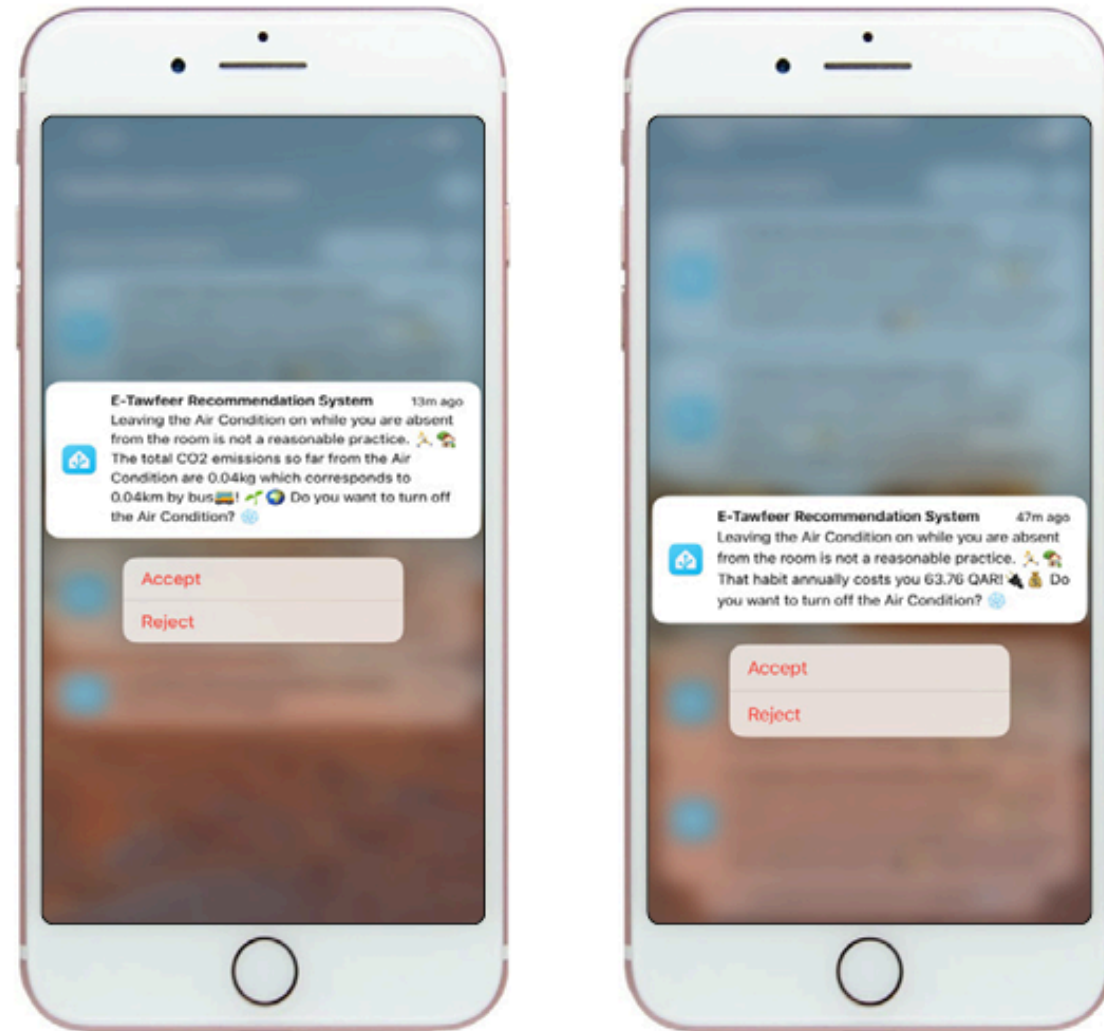
Testing Accuracy

94.55%

F1-score

- k-NN, Random Forest (RF), and MLP consistently yielded the highest accuracy and F1-score across all five datasets tested.
- RF algorithm was selected for edge deployment — trained locally on Raspberry Pi and imported into Home-Assistant.
- Confusion matrix: 748 true negatives, 598 true positives, 37 false positives, 32 false negatives — very low error rates.
- Achieved 95.12% testing accuracy and 94.55% F1-score, confirming RF is highly reliable for real-time edge-based occupancy detection.

Recommender System Results



80%

Positive User Response (mini-pilot study)

43%

Decrease in Lighting Power Usage (recommended vs. non-recommended weeks)

- Office-lighting usage averaged 3.78 hrs/day (Week 1) and 5.03 hrs/day (Week 4) without recommendations.
- With recommendations accepted: 2.98 hrs/day (Week 2) and 2.04 hrs/day (Week 3) — substantial reduction.
- 80% positive user response rate in the mini-pilot study.
- Recommendations combine ecological (CO₂/environmental) and economical (cost-savings) persuasion messages for higher engagement.

Conclusion

- DT system combining occupancy detection and energy-conservation recommender was implemented and validated over a ten-day lab study.
- Occupancy detection achieved 95.12% accuracy and 94.55% F1-score, confirming high reliability.
- Recommender mini-pilot revealed 80% positive user response with measurable energy savings.
- Results demonstrate tangible benefits of integrating ML, IoT, and recommender systems for building energy efficiency.

Future Work

- Integrate robust data-encryption techniques during transmission and storage for enhanced security.
- Employ decryption mechanisms at the edge to enable secure data analysis while preserving privacy.
- Explore alternative open-source platforms beyond Home-Assistant for broader applicability.
- Test the system in diverse environments including industrial facilities, community housing, and educational institutions.

References

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THANK YOU

